Chulalongkorn University · Department of Chemical Engineering

Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator

Document register and engineering queries

NDFH-V1-PM-DOC-001 · Rev 0 · Issued for teaching

Rev	Date	Description	Originator	Checker	Approver
0	25-Oct-26	Issued for teaching	A. Soottitantawat	—	—

Title block	
Client	Chulalongkorn University · Department of Chemical Engineering
Project	Natural Draft Fired Heater Ver 1 · H-100 Operator Training Simulator
Document no.	NDFH-V1-PM-DOC-001
Title	Document register and engineering queries
Drawings	PFD-100 rev 1; P&ID-001 rev C (screen drawing); NDFH-V1-PID-001 rev E (EPC alternative)
Scope	Every deliverable of the H-100 teaching project and the engineering queries open against them.

All numbers in this document come from the Natural Draft Fired Heater Ver 1 simulator model (design case and model runs) or are stated as assumed. It is a teaching document, not a design for construction.

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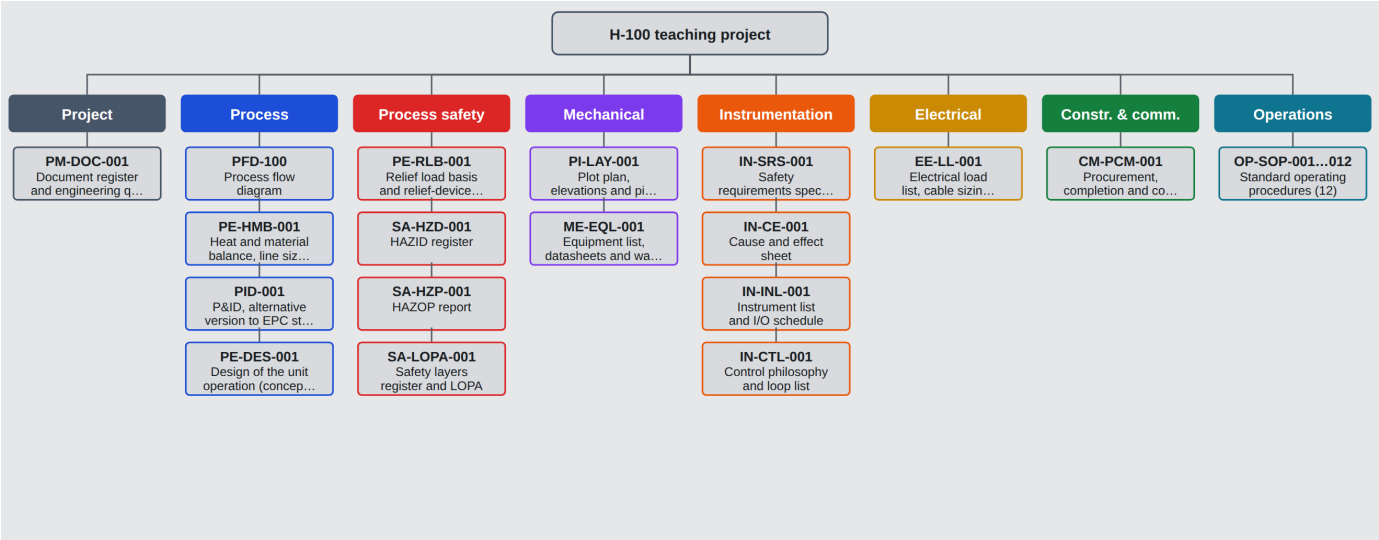
1 · Purpose

This register lists the documents of the H-100 teaching project: number, title, discipline, revision, status and where each one lives on the simulator. Status codes: IFD issued for design, IFR issued for review, IFT issued for teaching. The documents were written from the simulator model, the P&ID and the Design tab; where two documents disagree an engineering query (EQ) is raised in section 4.

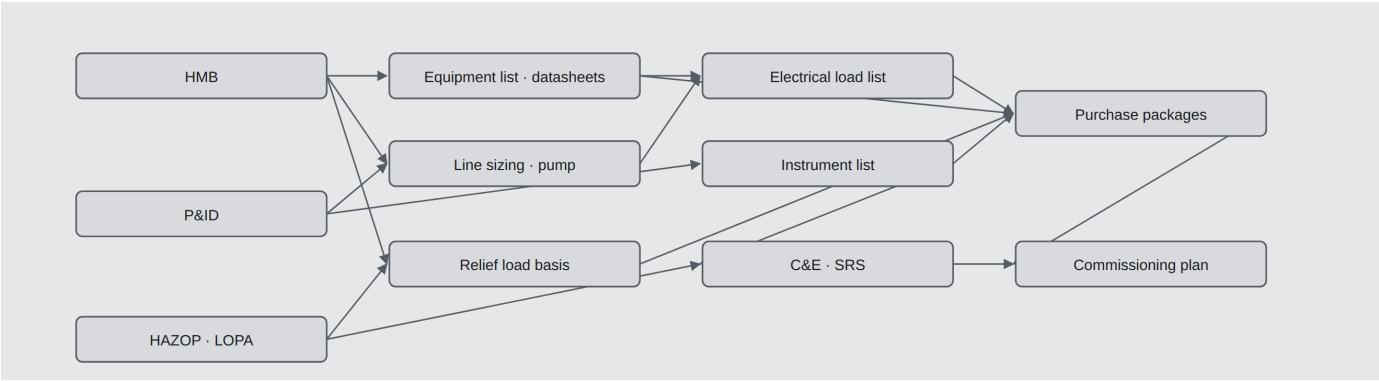
2 · Document register

Document no.	Title	Discipline	Rev	Status	Where it lives	Used for
NDFH-V1-PFD-100	Process flow diagram	Process	1	IFD	PFD tab · pfd-100.pdf	Basis of the P&ID, HAZID and the HMB
NDFH-V1-PE-HMB-001	Heat and material balance, line sizing and pump hydraulics	Process	0	IFT	Process engineering tab · NDFH-V1-PE-HMB-001.pdf	Every equipment and line size
NDFH-V1-PID-001	P&ID, alternative version to EPC standard (rev E)	Process	E	IFD	P&ID tab · NDFH-V1-PID-001.pdf	Instrument list, line list, layout, HAZOP
NDFH-V1-PE-DES-001	Design of the unit operation (conceptual sizing)	Process	0	IFT	Design of the unit operation tab	Heater datasheet
NDFH-V1-PI-LAY-001	Plot plan, elevations and piping isometrics (generated layout)	Mechanical	0	IFT	Layout & isometrics tab · Visual plant 3D tab	3-D plant, the outside-operator round, access and escape
NDFH-V1-ME-EQL-001	Equipment list, datasheets and wall-thickness checks	Mechanical	0	IFT	Mechanical tab · NDFH-V1-ME-EQL-001.pdf	Requisitions, layout, load list
NDFH-V1-PE-RLB-001	Relief load basis and relief-device sizing	Process safety	0	IFT	Relief & vent tab · NDFH-V1-PE-RLB-001.pdf	PSV datasheets, flare/drain loads
NDFH-V1-SA-HZD-001	HAZID register	Process safety	0	IFT	HAZID tab	HAZOP scope, layout
NDFH-V1-SA-HZP-001	HAZOP report	Process safety	A	IFR	HAZOP tab · NDFH-V1-SA-HZP-001.pdf	LOPA, SRS, recommendations
NDFH-V1-SA-LOPA-001	Safety layers register and LOPA	Process safety	0	IFT	Safety layers and LOPA tabs	SIL targets
NDFH-V1-IN-SRS-001	Safety requirements specification of the SIFs	Instrumentation	R0	IFT	SRS of SIF tab	SIS design, proof tests
NDFH-V1-IN-CE-001	Cause and effect sheet	Instrumentation	R0	IFT	C&E matrix tab · NDFH-V1-IN-CE-001.pdf	SIS configuration, function tests
NDFH-V1-IN-INL-001	Instrument list and I/O schedule	Instrumentation	0	IFT	Instrumentation & control tab · NDFH-V1-IN-INL-001.pdf	Instrument requisitions, I/O, loop checks
NDFH-V1-IN-CTL-001	Control philosophy and loop list	Instrumentation	0	IFT	Control loop design tab	DCS configuration
NDFH-V1-EE-LL-001	Electrical load list, cable sizing and hazardous-area schedule	Electrical	0	IFT	Electrical tab · NDFH-V1-EE-LL-001.pdf	Switchboard, UPS, cable schedule
NDFH-V1-CM-PCM-001	Procurement, completion and commissioning plan	Construction & commissioning	0	IFT	Procurement to commissioning tab · NDFH-V1-CM-PCM-001.pdf	Schedule, test plans, handover
NDFH-V1-OP-SOP-001…012	Standard operating procedures (12)	Operations	0	IFT	SOP tab	Training, start-up
NDFH-V1-PM-DOC-001	Document register and engineering queries	Project	0	IFT	This tab · NDFH-V1-PM-DOC-001.pdf	Document control

3 · Deliverable tree and dependencies



A change travels along the arrows below; that is why a change goes through management of change and every document carries a revision.



4 · Engineering queries

Fixed queries come from the studies (HAZOP, Safety layers, SRS); the others were raised automatically by checks that failed on the Engineering tabs. All are OPEN.

EQ	Raised by	Query	Proposed answer	Answer by	Status
EQ-01	HAZOP / Safety layers	FT-008 feeds FIC-003, FIAL-008 and FSLL-008: one high-reading failure blinds all three.	Give FSLL-008 its own transmitter FT-008B (LOPA L1 recommendation).	Instrumentation	OPEN
EQ-02	SRS of SIF	SIF-01 and SIF-03 response times are longer than half the process safety time; SIF-04 acts only at flame-out.	Shorter trip delays and faster valves; low-O_2 / high-CO trip (SRS findings).	Instrumentation	OPEN
EQ-03	Relief & vent	No coil relief valve; Safety layers lists PSV-102 as "confirm downstream relief covers it".	Relief study to decide: PSV-102 on the outlet, or credit FSLL-008 under API 521 with a documented SIF.	Process safety	OPEN
EQ-04	Relief & vent	E-100 shell design pressure is below 10/13 of the tube side: the tube-rupture case applies to the shell (Unit A side).	Add PSV-103 on the E-100 shell or raise the shell design pressure to 1308 kPa(g); agree with Unit A.	Process safety	OPEN
EQ-05	Mechanical	E-100 area rests on an assumed U and hot-fluid return temperature (A4, A5).	Vendor thermal rating; confirm the Unit A hot-oil conditions.	Mechanical	OPEN
EQ-06	Electrical	Feed and fuel properties (flash point, auto-ignition) are assumed (A7, A8).	Obtain the fluid data sheets before the area classification is issued.	Electrical	OPEN
EQ-07	Layout & isometrics	The heating-fluid return line (3"-HF-402 on the isometric) has no line number on the P&ID.	Add the line number to NDFH-V1-PID-001 at the next revision.	Process	OPEN
EQ-08	Layout & isometrics	The convection and stack platform has one ladder only: no second way down.	Add a second ladder or a stair from the opposite side of the platform.	Mechanical	OPEN
EQ-09	Process engineering (check)	Line 3"-HF-401 within its criteria at the maximum case: 2.82 m/s, 84.2 kPa/100 m vs ≤ 3 m/s, ≤ 45 kPa/100 m	Resolve before issue for construction.	Process	OPEN
EQ-10	Mechanical (check)	Tube skin alarm TI-024 below the design metal temperature: 430 °C vs 425 °C	Resolve before issue for construction.	Mechanical	OPEN

5 · Checks

Check	This document	Compared with	Result
Every register entry points to a tab or a file that exists	18 documents	tabs and files in this page	✓
Every failed check on the Engineering tabs has an engineering query	3 failed checks	listed in section 5	✓